DPP No: 13

Maximum Time 50 Min

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C³

(COMPETISHUN CRASH COURSE)

PHYSICS

TARGET JEE-MAINS

SYLLABUS : EMW AND PRINCIPLE OF COMMUNICATION

1. The electromagnetic waves used in the telecommunication are

(A) ultraviolet (B) infra-red (C) visible (D) microwaves.

2. If an electromagnetic wave propagating through vacuum is described by

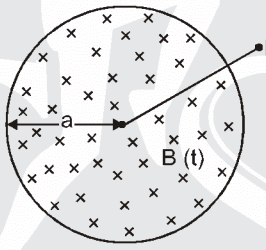
$E = E_0 \sin(kx - \omega t)$; $B = B_0 \sin(kx - \omega t)$,

(A) $E_0 k = B_0 \omega$ (B) $E_0 B_0 = \omega k$ (C) $E_0 \omega = B_0 k$ (D) $E_0 B_0 = \omega / k$

3. Which of the following is independent of wavelength ?

(A) k (B) ω (C) ωk (D) k / ω

4. A uniform but time varying magnetic field B (t) exists in a circular region of radius a is directed into the plane of the paper as shown. The magnitude of the induced electric field at point P at a distance r from the centre of the circular region.



(A) is zero (B) decreases as 1/r

(C) increases as r (D) decreases as $1/r^2$

5. An accelerated electric charge emits

(A) β – rays (B) γ – rays (C) e.m. waves (D) none of the above

6. Electromagnetic waves in nature are

(A) longitudinal (B) longitudinal stationary

(C) transverse (D) transverse – stationary

7. The speed of e. m. waves is given by the relation

(A) $\mu_0 \varepsilon_0$ (B) $\sqrt{\mu_0 \varepsilon_0}$ (C) $1/\mu_0 \varepsilon_0$ (D) $1/\sqrt{(\mu_0 \varepsilon_0)}$

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8. In an electromagnetic wave, electric field **E** and magnetic field **B** are

(A) mutually perpendicular to each other (B) all parallel

(C) at 30^0 to each other (D) at 60^0 to each other

9. If **E** and **B** be the electric and magnetic fields of electromagnetic waves, then the direction of propagation of e. m. wave is along the direction of

(A) E (B) B (C) $E \times B$ (D) None of the above

10. Visible range of wavelength in cm is

(A) 3×10^{-6} to 10^{-10} (B) 7×10^{-5} to 4×10^{-5}

(C) 4×10^{-5} to 3×10^{-6} (D) 6×10^{4} to 1.5×10^{3}

11. The shortest wavelength is for

(A) γ – rays (B) x – rays (C) ultraviolet rays (D) microwaves

12. Which has the longest wavelength `

(A) infrared light (B) ultraviolet light (C) radio–waves (D) X–rays

13. The upper atmosphere layer is known as

(A) troposphere (B) mesosphere (C) ionospere (D) chromosphere

14. Ionosphere is mainly composed of

(A) electrons and positive ions (B) ozone gas

(C) nitrogen and oxygen (D) b and c both

15. Finger prints on a piece of paper may be detected by sprinkling flourescent powder on the paper and then looking it into

(A) dark-light (B) sun-light (C) Infra-red-light (D) ultraviolet light

16. Ozone layer in atmosphere exist at the height of

(A) 20 km (B) 50 km (C) 120 km (D) 150 km

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17. Ozone layer protectes the living cells from

(A) ultraviolet radiations (B) unfra –red radiations

(C) X–rays (D) all the radiations

18. The ionosphere does not allow to pass the waves which are termed as

(A) microwaves (B) visible light waves

(C) 1 and 2 both (D) amplitude modulated waves

19. The ionosphere bends the e. m. waves having the frequencies

(A) less than 40 MHz (B) beyond 40 MHz

(C) nothing is certain (D) depends on the moisture present

20. The waves which can travel directly along surface of the earth are known as

(A) ground waves (B) X–rays (C) α–rays (D) sky waves

21. The number of radio frequency carrier waves transmitted by a television transmitter is

(A) three (B) two (C) one (D) four

22. The S.I unit of displacement current is

(A) H (B) A (C) Fm^{-1} (D) C

23. An electromagnetic radiation of frequency ν, wavelength λ, travelling with velocity c in air, enters a glass slab of refractive index μ. The frequency, wavelength and velocity of light in the glass slab will be respectively :

(A) $\frac{\nu}{\mu}, \frac{\lambda}{\mu}$ and $\frac{c}{\mu}$ (B) $\nu, \frac{\lambda}{\mu}$ and $\frac{c}{\mu}$ (C) $\nu, 2\lambda$ and $\frac{c}{\mu}$ (D) $\frac{2\nu}{\mu}, \frac{\lambda}{\mu}$ and c

24. The speed of electromagnetic waves is independent of

(A) wavelength (B) frequency

(C) intensity (D) medium, in which it travels

25. If ε_0 and μ_0 are the electric permittivity and magnetic permeability in free space, ε and μ are the corresponding quantities in a medium, then index of refraction of the medium is

(A) $\sqrt{\frac{\varepsilon_0\mu}{\varepsilon\mu_0}}$ (B) $\sqrt{\frac{\varepsilon}{\varepsilon_0}}$ (C) $\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}}$ (D) $\sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}}$

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ANSWER KEY

1.	(D)	**2.**	(A)	**3.**	(D)	**4.**	(B)	**5.**	(C)
6.	(C)	**7.**	(D)	**8.**	(A)	**9.**	(C)	**10.**	(B)
11.	(A)	**12.**	(C)	**13.**	(C)	**14.**	(A)	**15.**	(D)
16.	(B)	**17.**	(A)	**18.**	(D)	**19.**	(B)	**20.**	(A)
21.	(B)	**22.**	(B)	**23.**	(B)	**24.**	(C)	**25.**	(D)